

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	D	F	H	S_1
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$ $+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{1}{2}F_{r_1}{}^T C F_{r_1} M_F$ $+\overline{L_{a_1,i_1}} P_R F_{r_1} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y^1,a_1,j_1}$ $+\overline{y_{2,i_1,r_1}} F_{r_1}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} F_{r_1}{}^T C P_L L_{a_1,i_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1} S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}} P_R F_{r_1} y_{2,i_1,r_1} \mathcal{I}_{y^2,a_1,j_1} S_{2,j_1}^\dagger$	$+\overline{\lambda_{\text{T3}} S_{2,j_2}} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{\text{T3},a_1,a_2,j_2,j_1} \lambda_{\text{T3}} S_{2,j_1}^\dagger$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H1}}^1$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H1}}^2$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H2}}^1$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H2}}^2$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$ $+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\frac{1}{2}F_{j_1,r_1}{}^T C F_{j_1,r_1} M_F$ $+\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y^1,a_1,j_1,j_1}$ $+\overline{y_{2,i_1,r_1}} F_{j_1,r_1}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} F_{j_1,r_1}{}^T C P_L L_{a_1,i_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1,j_1} S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y^2,a_1,j_1,j_1} S_{2,j_1}^\dagger$	$+\overline{\lambda_{\text{T3}} S_{2,j_2}} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{\text{T3},a_1,a_2,j_2,j_1} \lambda_{\text{T3}} S_{2,j_1}^\dagger$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H1}}^1$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H1}}^2$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H2}}^1$ $+H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H2}}^2$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$ $+\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1,j_1}$ $+\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y^2,a_1,j_1,j_1} S_{2,j_1}^\dagger$	$+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H2}}^1$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H2}}^1$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H2}}^2$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H2}}^2$	$-S_1 S_1^\dagger M_{S_1}^2$ $+S_1 S_{2,j_2} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1$ $+\frac{1}{2} S_1^2 (S_1^\dagger)^2 \lambda_{S_1}^1$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$ $+\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y^1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} L_{a_1,i_1}{}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1,j_1} S_{1,j_1}^\dagger$	$+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H1}}^1$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_2,j_1}^1 \lambda_{\text{H1}}^1$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H1}}^2$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_2,j_1}^2 \lambda_{\text{H1}}^2$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$ $+D_{\mu_1}(S_{2,j_1})^\dagger D_{\mu_1}(S_{2,j_1})$	$-\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F$ $+\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y^1,a_1,j_1,j_1}$ $+\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} L_{a_1,i_1}{}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1,j_1} S_{1,j_1}^\dagger$ $+\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y^2,a_1,j_1,j_1} S_{2,j_1}^\dagger$	$+\overline{\lambda_{\text{T3}} S_{2,j_2}} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{\text{T3},a_1,a_2,j_2,j_1} \lambda_{\text{T3}} S_{2,j_1}^\dagger$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H1}}^1$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H1}}^1$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_2,j_1}^2 \lambda_{\text{H1}}^2$ $+\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{\text{H1},a_1,a_2,j_2,j_1}^2 \lambda_{\text{H1}}^2$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H2}}^1$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^1 \lambda_{\text{H2}}^1$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H2}}^2$ $+\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{\text{H2},a_1,a_2,j_1,j_2}^2 \lambda_{\text{H2}}^2$	

Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	constant	F	D	S_1	D
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \end{aligned}$	$+\frac{1}{2} i F_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (F_{r_1})$			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \end{aligned}$	$+\frac{1}{2} i F_{j_1,r_1}^T C \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$		$\begin{aligned} & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^2 \lambda_{S2}^2 \end{aligned}$	$+i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$		$+D_{\mu_1} (S_1)^\dagger D_{\mu_1} (S_1)$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\begin{aligned} & -M_{S_2}^2 S_2 S_2^\dagger \\ & +S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1 \\ & +\frac{1}{2} S_2^2 (S_2^\dagger)^2 \lambda_{S2}^1 \end{aligned}$	$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^2 \lambda_{S1}^2 \end{aligned}$	$+i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		$\begin{aligned} & -S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2 \\ & -M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2 \\ & +S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^1 \lambda_{S1}^1 \\ & +\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S1,j_1,j_3,j_2,j_4}^2 \lambda_{S1}^2 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^1 \lambda_{S2}^1 \\ & +\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S2,j_1,j_3,j_2,j_4}^2 \lambda_{S2}^2 \end{aligned}$	$+i \overline{F_{j_1,r_1}} \gamma^{\mu_1} D_{\mu_1} (F_{j_1,r_1})$			

Field combinations 3 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	H	S_2	D	S_2	F
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0						
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0						
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_1 y_{1,i_1,r_1} \mathcal{I}_{\gamma^1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} L_{a_1,i_1}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1} S_1^\dagger$	$+\overline{\lambda_{T3}} S_{2,j_1} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger$ $+H_{a_1} H_{a_2} S_1 \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_{T3} S_{2,j_1}^\dagger$ $+H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{H1,a_1,a_2}^1 \lambda_{H1}^1$				
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$			$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$		$+\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1,j_1}$ $+\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{\gamma^2,a_1,j_1} S_2^\dagger$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$						

Field combinations 4 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+H_{a_1} H_{a_2} S_{1,j_1} \mathcal{L}_{T3,a_1,a_2,j_1} \lambda_{T3} S_2^\dagger$ $+ \overline{\lambda_{T3}} S_2 \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+ H_{a_2} S_2 H_{a_1}^\dagger S_2^\dagger \mathcal{L}_{H2,a_1,a_2}{}^1 \lambda_{H2}{}^1$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	